UrbanFlow AI

AI-Powered Smart Traffic Intelligence System

An intelligent traffic monitoring system that detects vehicles and analyzes traffic density.

Run Traffic Analysis

Click 'Run Traffic Analysis' to start AI detection.

UrbanFlow AI | Computer Vision & Traffic Analytics

Help agents write better apps

Install the official Streamlit skills so AI coding agents can build and debug your apps.

Install Don't show again

ℹ️ About UrbanFlow AI

An AI-powered traffic monitoring and analytics system using YOLO11.

🚦 UrbanFlow AI

AI-Powered Smart Traffic Intelligence System

An intelligent traffic monitoring system that detects vehicles and analyzes traffic density.

📹 Upload Traffic Video

Choose a traffic video

Upload 200MB per file • MP4, AVI, MOV, MKV

Run Traffic Analysis

UrbanFlow-AI

Public

Pin | Watch 0 | Fork 0 | Star 0

main | 1 Branch | 0 Tags | Go to file T | Add file | Code

RadhaRani07 Initial commit: UrbanFlow AI — 41f3164 · 7 minutes ago — 1 Commit

.gitignore	Initial commit: UrbanFlow AI	7 minutes ago
app.py	Initial commit: UrbanFlow AI	7 minutes ago
detector.py	Initial commit: UrbanFlow AI	7 minutes ago
requirements.txt	Initial commit: UrbanFlow AI	7 minutes ago

README

Add a README

Help people interested in this repository understand your project.

Add a README

About

AI-powered Smart Traffic Monitoring and Analytics System using YOLO11, OpenCV and Streamlit.

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors 1

RadhaRani07

main 1 Branch 0 Tags Go to file T Add file Code

RadhaRani07 Update README.md 234ef2b · 1 minute ago 3 Commits

.gitignore	Initial commit: UrbanFlow AI	22 minutes ago
README.md	Update README.md	1 minute ago
app.py	Initial commit: UrbanFlow AI	22 minutes ago
detector.py	Initial commit: UrbanFlow AI	22 minutes ago
requirements.txt	Initial commit: UrbanFlow AI	22 minutes ago

README

UrbanFlow AI

AI-Powered Smart Traffic Monitoring and Analytics System

UrbanFlow AI is a student-built traffic monitoring and analytics project that uses computer vision to detect and track vehicles in uploaded traffic videos.

The system uses YOLO11 for vehicle detection, ByteTrack for tracking, OpenCV for video processing, and Streamlit for

Deploy

UrbanFlow AI

Smart Traffic Monitoring System

Navigation

- Dashboard
- Traffic Analysis
- Analysis History
- About Project

AI-Powered Traffic Analytics

UrbanFlow AI

Smart Traffic Intelligence Dashboard

Welcome to UrbanFlow AI — an AI-powered system for vehicle detection, tracking and traffic analytics.

Project Overview

Vehicle Detection

YOLO11-powered vehicle detection and tracking.

Traffic Analytics

Vehicle counting and traffic density classification.

Analysis Reports

Download traffic analysis reports in CSV format.

Analysis History

Access previously saved traffic analysis records.

Deploy

UrbanFlow AI

Smart Traffic Monitoring System

Navigation

- Dashboard
- Traffic Analysis
- Analysis History
- About Project

AI-Powered Traffic Analytics

Technology Stack

Python | YOLO11 | OpenCV | Streamlit | Pandas

Latest Traffic Analysis

Total Vehicles	Cars	Motorcycles
194	36	120

Buses	Trucks
13	25

Latest Traffic Density

Traffic Density: HIGH

UrbanFlow AI | Computer Vision & Traffic Analytics

Deploy

UrbanFlow AI

Smart Traffic Monitoring System

Navigation

- Dashboard
- Traffic Analysis
- Analysis History
- About Project

AI-Powered Traffic Analytics

Traffic Analysis

UrbanFlow AI

AI-Powered Smart Traffic Intelligence System

An intelligent traffic monitoring system that detects vehicles and analyzes traffic density.

Upload Traffic Video

Choose a traffic video

Upload 200MB per file • MP4, AVI, MOV, MKV

Run Traffic Analysis

Click 'Run Traffic Analysis' to start AI detection.

Deploy

UrbanFlow AI

Smart Traffic Monitoring System

Navigation

- Dashboard
- Traffic Analysis
- Analysis History
- About Project

AI-Powered Traffic Analytics

Analysis History

Date & Time	Total Vehicles	Cars	Motorcycles	Buses	Trucks	Traffic Density	Average Confidence (%)
2026-10-31 14:02:28	194	36	120	13	25	HIGH	49.1
2026-10-31 14:11:20	194	36	120	13	25	HIGH	49.1
2026-10-31 14:17:39	194	36	120	13	25	HIGH	49.1
2026-10-31 14:29:20	194	36	120	13	25	HIGH	49.1

Download Analysis History

Deploy

UrbanFlow AI

Smart Traffic Monitoring System

Navigation

- Dashboard
- Traffic Analysis
- Analysis History
- About Project

AI-Powered Traffic Analytics

ℹ️ About UrbanFlow AI

UrbanFlow AI is an AI-powered Smart Traffic Monitoring and Analytics System.

🎯 Project Objective

The objective is to detect, track and analyze vehicles from traffic videos using Computer Vision.

🛠️ Technologies Used

- Python
- YOLO11
- OpenCV
- Streamlit
- Pandas

✨ Key Features

- Vehicle Detection and Tracking
- Vehicle Counting
- Traffic Density Classification
- Traffic Analytics Dashboard

Deploy

UrbanFlow AI

Smart Traffic Monitoring System

Navigation

- Dashboard
- Traffic Analysis
- Analysis History
- About Project

AI-Powered Traffic Analytics

- Python
- YOLO11
- OpenCV
- Streamlit
- Pandas

✨ Key Features

- Vehicle Detection and Tracking
- Vehicle Counting
- Traffic Density Classification
- Traffic Analytics Dashboard
- Analysis History and CSV Reports
- AI Detection Confidence

📌 Project Type

Computer Vision | Artificial Intelligence | Machine Learning